Hong Kong Institute of Vocational Education

Department of Information and Communications Technology

Final Year Project (ITP4913M)

Project Proposal

Please read the instructions in the Guidelines for Writing Project Proposal before completing this form.

Student Name:	Student ID(s):

Supervisor:

Project Title: Smart Parking – Easy In Easy Out

Statement of problem to be solved:

1. Crowded parking area let driver car park difficulty.
2. Shopping mall losing the potential customer due to long queue time.
3. Payment inconvenient, mostly support octopus or cash only.
4. Lower Security, depend human security to monitor the parking is not reliable.

Background to the problem:

1. Hong Kong densely populated, in cases of not enough parking area supply.

2. The old parking system is not effective for existing parking demand and leads to long queue time.

3. Aims to follow smart city scheme and keep up with nearly country smart city development, the traditional parking payment method should be changed to become fast and user-friendly.

Outline of proposed solution:

The Smart Parking System will:	1. Issue Automatic identification of license plates 2. Issue function to locate vehicle location in car park 3. Issue Quickpass payment method 4. Issue Available parking space monitoring 5. Customer usage analysis
Milestone	Milestone 1 - Centralized database and system architecture Milestone 2 - Automatic identification of license plates Milestone 3 – Quickpass payment functions Milestone 4 - Retrieve vehicle function Milestone 5 - Customer usage analysis

Problem: Hong Kong densely populated, in cases of not enough parking area supply.
Solution: A complete database and different hardware to monitor the usage situation in car park. Provide accurate empty car park information for driver who are seeking parking service.

Problem: The old parking system is not effective for existing parking demand and leads to long queue time.
Solution: To change the process of car in and car out during car park. Using Automatic identification of license plates function to reduce the waiting time.

Problem: Aims to follow smart city scheme and keep up with nearly country smart city development, the traditional parking payment method should be changed to become fast and user-friendly.
Solution: Using electronic settlement (PayPal), provide Quickpass payment. Everything which is related the car parking can control on the mobile apps such as car park finding, vehicle recovery and parking fee payment.

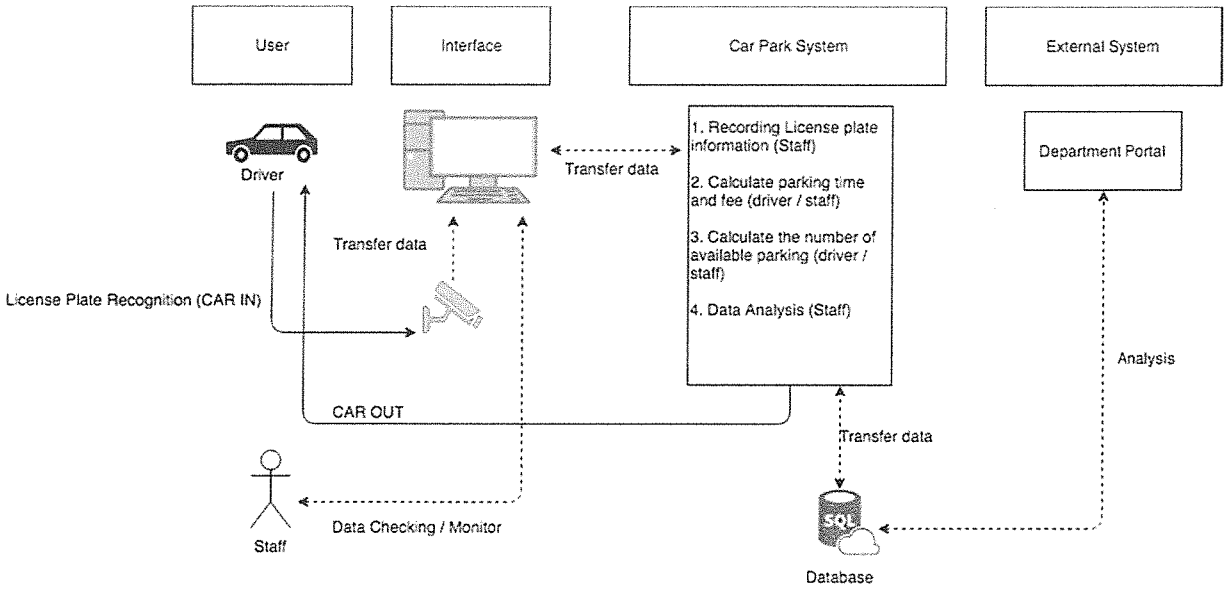


Figure 1 Diagram for high level system overview

Explanation of why proposed solution is appropriate:

To develop a new payment method and improve traditional parking system in Hong Kong. For example, we adopt the mobile payments to replace traditional payment method and provide car finding system in parking. It can improve the queue time during parking. In the same time, it is advantage for the parking diversion in Hong Kong densely populated.

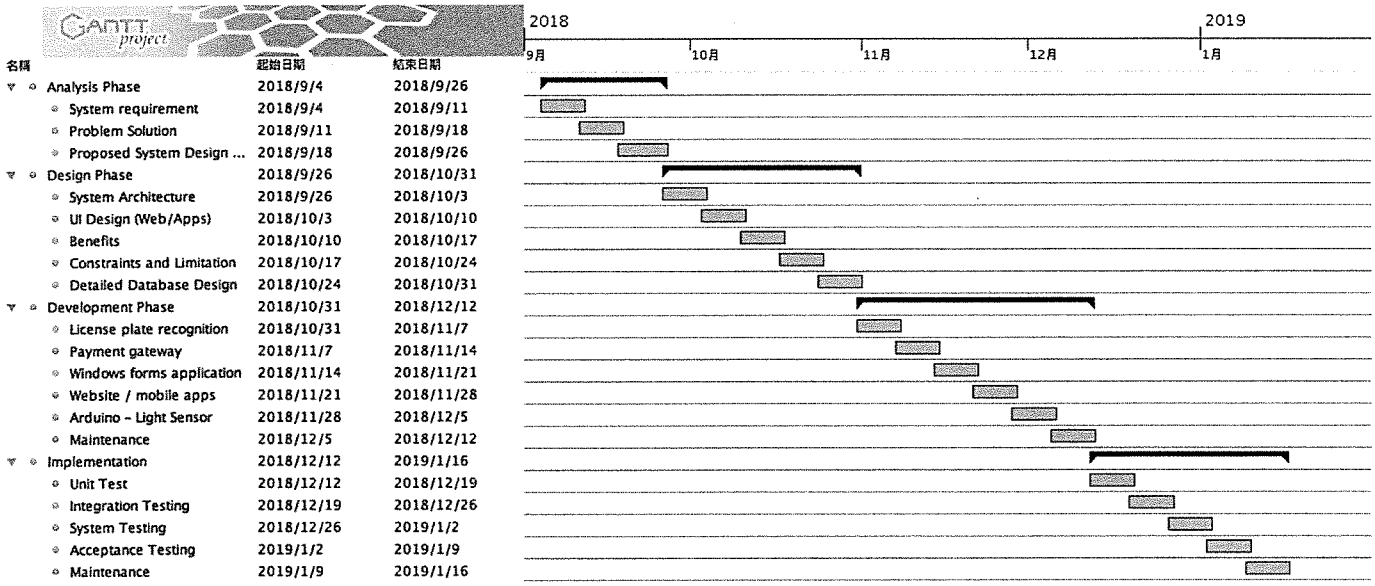
1. License plate recognition is a part of smart parking system. It is good for record the empty parking space and calculate the parking time for each driver. The data will collect for car park payment reference.

2. To change the payment method via by electronic settlement (PayPal). The driver no need to show out anything such as Octopus or cash when they leave. It is good for reduce car park waiting time indirectly.

3. The mobile apps will provide the search function for located the car park which is near. The system will process the payment when the driver leaves, and car finding service in the car park. It is good to monitor the traffic flow and maximize the Available parking space when the driver easies find their car in the car park and leave car park quickly. It is easy in and easies out. To avoid losing the potential customer due to the long queue.

Main stages involved in the project, with estimated time to completion of each stage:

This is a tentative time plan. It will be modified by case.



名稱	起始日期	結束日期
Analysis Phase	2018/9/4	2018/9/26
System requirement	2018/9/4	2018/9/11
Problem Solution	2018/9/11	2018/9/18
Proposed System Design ...	2018/9/18	2018/9/26
Design Phase	2018/9/26	2018/10/31
System Architecture	2018/9/26	2018/10/3
UI Design (Web/Apps)	2018/10/3	2018/10/10
Benefits	2018/10/10	2018/10/17
Constraints and Limitation	2018/10/17	2018/10/24
Detailed Database Design	2018/10/24	2018/10/31
Development Phase	2018/10/31	2018/12/12
License plate recognition	2018/10/31	2018/11/7
Payment gateway	2018/11/7	2018/11/14
Windows forms application	2018/11/14	2018/11/21
Website / mobile apps	2018/11/21	2018/11/28
Arduino - Light Sensor	2018/11/28	2018/12/5
Maintenance	2018/12/5	2018/12/12
Implementation	2018/12/12	2019/1/16
Unit Test	2018/12/12	2018/12/19
Integration Testing	2018/12/19	2018/12/26
System Testing	2018/12/26	2019/1/2
Acceptance Testing	2019/1/2	2019/1/9
Maintenance	2019/1/9	2019/1/16

Main Deliverables:

1. System analysis and design
2. User interface
3. Software open source code
4. implementation plan result
5. System spec
6. User guideline

The (tentative) responsibilities of each member:

Name:	**Responsible:**
Student A	License plate recognition, UI design, Database design
Student B	Light Sensor, UI design, Database design
Student C	Payment gateway, Website design, Database design
Student D	Mobile apps design, Database design

Projected Resource Requirements:

Hardware (Computer/any specialized hardware):

1. Camera
2. Computer
3. Mobile phone
4. Light sensor

Software:

Flutter, Photoshop, XCode, Drupal, Firebase, Dialogflow, Anaconda, Visual Studio, Android studio

Owner and location of resources:

1. Light Sensor – VTC
2. Mobile phone – Groupmate
3. Computer – Groupmate
4. Camera – Groupmate

External involvement:

1. OpenCV: https://anaconda.org/conda-forge/opencv
2. Payment Gateway: https://stripe.com/us/payments
3. Firebase: https://firebase.google.com/
4. Flutter: https://flutter.io/
5. Dialogflow: https://dialogflow.com/